Cradle-to-Gate Carbon Footprint: Vestas V236-15MW Offshore Turbine

Life-cycle assessment to ISO 14040 and 14044 for a single turbine within a 990 MW offshore wind plant.

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THE QUESTION

Wind power is treated as emissions-free because it produces no emissions in operation. What does it cost to build one before it generates anything?

1 The emissions sit in manufacturing, and mostly in steel

Foundations contribute 25% of cradle-to-gate global warming potential and the tower 13%. Steel is roughly 82.9% of total turbine mass, so the footprint is largely a steel procurement question rather than a turbine design question.

2 Installation is small but not nothing

The set-up stage contributes about 4%, from marine gas oil burned by service and installation vessels. It is a measurable component worth managing, which is easy to miss when the manufacturing figure dominates the chart.

3 Ordinary fuel combustion is the mechanism, not exotic industrial gases

Carbon dioxide accounts for 92% of the impact and methane 6%. Sulphur hexafluoride, despite an extremely high global warming potential per unit, contributes about 1% because so little is used. Intuition about which substances matter is misleading here.

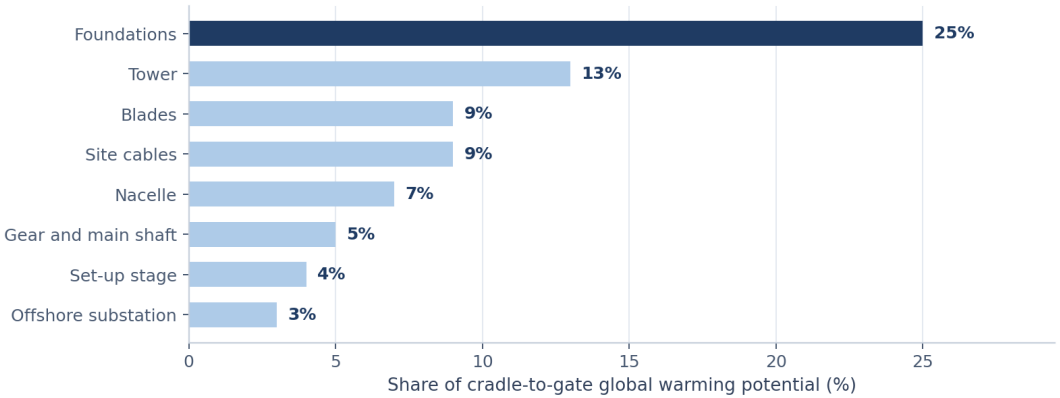
4 Benchmarking gave the finding a reference point

Results were compared against Siemens Gamesa, the direct competitor in the offshore segment, so the reduction levers proposed (low-emission steel, higher recyclable content, transport optimisation) are aimed at the stages where the gap is real.

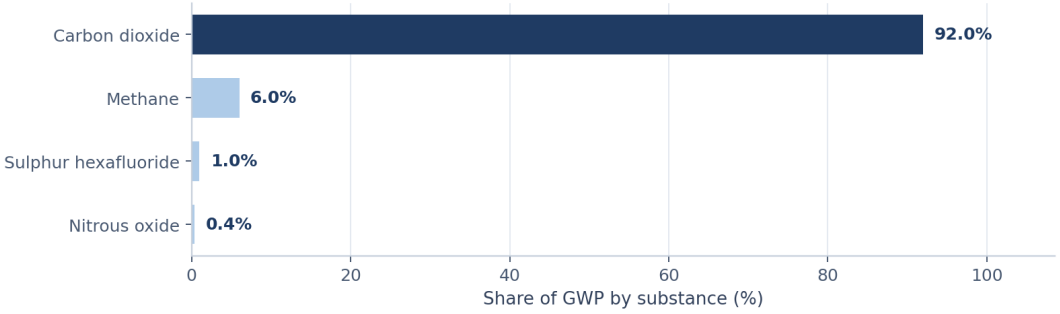
LIMITATIONS, AND WHAT WOULD CHANGE THE ANSWER

The assessment is cradle-to-gate by design, which means it deliberately excludes site operation, service logistics, end-of-life and repowering. It therefore says nothing about lifetime net carbon balance and should not be read as an argument for or against offshore wind. Emissions data is drawn from Vestas's own published LCA reporting rather than independently measured, so the study inherits both the manufacturer's boundary choices and its allocation assumptions.

Manufacturing dominates: the tower and foundations alone are 38%



Fuel combustion, not exotic gases, is the story



Built with Life-cycle assessment · ISO 14040 and 14044 · GWP100 · competitor benchmarking

Delivered material and stage inventory · impact quantification · competitor comparison · prioritised reduction levers